

 UNIVERSITY OF TECHNOLOGY - VNUHCM FACULTY OF COMPUTER SCIENCE AND ENGINEERING	HOMEWORK 2		Semester/Academic year		1	2026-2027
			Due		07/10/2026	
	Course title	Machine Learning				
	Course ID	CO3117				
	Duration	Take-home	Homework set		HW2	
Notes: Handwritten homework, submitted as homework/hw2-submission.pdf with tag hw2.						

Instructions

- **Decision trees** (Chapter 2). Five problems, each worth 2 points (0.5 per part); total 10 points.
- **Write by hand**, on paper or with a stylus. Show your working: a bare number earns no credit. Put your full name, student ID and the set number at the top of the first page.
- **Work on your own.** Do not use AI tools, solution manuals or classmates' work before your submission is tagged. You may use your lecture notes and the course slides. Any later AI use goes in AI_USE.md, as for the weekly drills.
- **Submit in your course repository.** Scan all pages, in order, into one PDF of at most 5 MB, named homework/hw2-submission.pdf. Commit it, run python tools/check.py --checkpoint hw2, then git tag hw2 and git push origin main hw2.
- **Deadline: 07/10/2026 07:00** (the time GitHub receives the tag hw2). Solutions are released on 07/10/2026. Afterwards you may add homework/hw2-corrections.md; never replace or re-tag the submission.

Problem 1 (L.O.2) (2 pts): Answer the following questions about impurity measures in decision trees. A node contains 14 training samples: 9 Positive and 5 Negative.

- Give the formula of the Gini index and compute it for this node.
- A test on attribute A sends 5 samples (4 Positive, 1 Negative) to the left child and 9 samples (5 Positive, 4 Negative) to the right child. Compute the weighted Gini index of the children and the reduction in Gini index.
- Compare entropy and the Gini index for a two-class node: where does each reach its maximum, what is the maximum value, and why is Gini sometimes preferred in practice?
- Which classic tree algorithm uses the Gini index by default, and what kind of splits does it produce?

Problem 2 (L.O.2) (2 pts): Answer the following questions about attribute selection. A data set has 10 samples (7 Yes, 3 No) and two attributes. Attribute A has two values: a_1 covers 5 Yes and 0 No, a_2 covers 2 Yes and 3 No. Attribute B has three values: b_1 covers 3 Yes and 0 No, b_2 covers 4 Yes and 1 No, b_3 covers 0 Yes and 2 No.

- Compute the information gain of attribute A .
- Compute the information gain of attribute B . Which attribute does ID3 choose for the root?
- Compute the split information and the gain ratio of A and of B . Which attribute is chosen by the lecture's rule (largest gain ratio)?
- Explain why information gain and gain ratio can disagree, and what problem of information gain the gain ratio is designed to correct.

Problem 3 (L.O.2) (2 pts): Answer the following questions about overfitting and pruning in decision trees:

- Why does a decision tree grown until every leaf is pure tend to overfit?
- Distinguish pre-pruning from post-pruning, and give one stopping rule used in pre-pruning.
- An internal node covers 20 validation examples. Its subtree misclassifies 6 of them; replacing the subtree by a leaf that predicts the majority class misclassifies 5. Should the subtree be pruned under reduced-error pruning? Explain.

- D. CART's cost-complexity pruning minimises $R_\alpha(T) = R(T) + \alpha|T|$, where $|T|$ is the number of leaves. Tree T_1 has error $R=0.10$ with 8 leaves, and tree T_2 has $R=0.14$ with 3 leaves. Which tree is chosen for $\alpha=0.01$ and for $\alpha=0.005$? For which α are they equally good?

Problem 4 (L.O.2) (2 pts): Answer the following questions about regression trees. A node contains six training examples with targets 2, 4, 6 (Region = North) and 12, 14, 16 (Region = South).

- A. What value does a regression-tree leaf predict?
- B. Compute the mean and the variance of the targets at this node.
- C. Split the node on Region. Compute the weighted variance of the children and the variance reduction.
- D. Why can entropy not be used as the splitting criterion for regression? Give one stopping criterion suited to regression trees.

Problem 5 (L.O.2) (2 pts): Answer the following questions about missing values and many-valued attributes in decision trees:

- A. How does C4.5 handle a training example whose value for the tested attribute is missing?
- B. Attribute Humidity is known for 9 of 10 examples: 6 are High and 3 are Normal. With the approach of part A, how is the tenth example sent down the two branches?
- C. Continue part B. Among the 9 examples with known Humidity, High covers 1 Yes and 5 No, and Normal covers 3 Yes and 0 No. C4.5 computes the gain on the examples with known values and multiplies it by the fraction of known values. Compute the gain of Humidity.
- D. A data set of 12 examples (7 Yes, 5 No) has an attribute CustomerID with a different value for every example. What information gain does it obtain, and what is its split information? Why should it not be chosen?

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